

12345678

A

cart-rp2040

File: cart-rp2040.kicad_sch

esp32s3-sd

File: esp32s3-sd.kicad_sch

usb-uart

File: usb-uart.kicad_sch

power

File: power.kicad_sch

B

C

FujiNet-Astrocade-Rev0 — all-in-one Bally Astrocade FujiNet cartridge.
RP2040 serves the 8K cart window (pico/astrocade fujicade firmware); 3x 74LVC245 level-shift the 5V bus. ESP32-S3-WROOM-1-N16R8 (fujiversal-astrocade) is USB host to the RP2040 (native USB, VID 0xCafe). Single USB-C via CP2102N flashes the S3; RP2040 flashes through the S3 (PICOBOOT doorbell / GPIO4-GPIO5 forcing).
Edge: 1x26 0.100in single-sided per ballyalley (Tilton/MCM). SW1 = top-face RESET.

D

E

F

Sheet: /
File: FujiNet-Astrocade-Rev0.kicad_sch

Title:

Size: A3Date:Rev:

KiCad E.D.A. 10.0.6Id: 1/5

12345678

FujiNet-Astrocade Rev0 cart bus. 26-contact single-sided edge (1x26, 0.100in),
pinout per Jay Tilton / MCM Design (ballyalley.com): 13 addr + 8 data + /CCS + 5V + 3x GND.
Console is read-only: U7 drives D0-7 only while /CCS low (OE gated by the 5V signal, LVC245 5V-tolerant).
U8/U9 translate 5V addr+enable down to 3V3 (always enabled, B->A).
Firmware map (astrocade_cart.h): GP0-12=A0-12, GP13=/ENABLE, GP14-21=D0-7,
GP22 selftest, GP25 LED, GP26 console-5V sense (100k/100k), GP27 debug TX.
SW1 RESET (top of cart) pulls RP RUN + S3 EN via D8. SW2+R20 = BOOTSEL (as Pico).
Q2/Q3: ESP32-S3 GPIO4/GPIO5 force RUN / QSPI_SS for PICOB00T recovery (as INTV Rev0).







